



# Methods

## Statistical analysis–Limited imaging

- Compared to MRI, ultrasound & EMA are limited in terms of the articulators that could be imaged:
  - Ultrasound: excluded /ɹ/, /m, n, ŋ/, /ɑ, ɒ, ɐ, æ, a, aʊ, aɪ, ə/ (56.37%)
  - EMA: excluded /m, n, ŋ/ (19.53%)





# Speaker normalization

- EMA/annotated MRI: by-speaker & by-dimension z-scoring of the coordinates.
- For the raw imaging data: Spatial Transformer Networks (which rotated/resized each frame).

